



The Industrialization of LLM training

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Poolside AI

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Grenoble Data Science meetup
May 21st 2026



A one-year long research workshop on large multilingual models and datasets



<https://bigscience.huggingface.co/>

Started May'21

1,000 researchers - 60 countries - 250 institutions

Central asset: BLOOM: 176B multilingual model

Created:

- Licenses
- Papers
- Datasets

- Community
- Self-confidence



Thomas Wolf
@Thom_Wolf



3 years ago we trained a big LLM together with +1000 people

one of the part I'm most proud of is how it became the school for a generation of model training engineers as well as a strong push for sharing and open-sourcing knowledge in the field



Matthias Gallé @mgalle · Aug 23, 2024

Packing for a weekend I found this.

It is hard to believe that @BigScienceLLM really happened. The first time I heard of the idea my take was "this is going to be fun... but not going to work"

Kudos to @Thom_Wolf for the vision



Where are all the models???



Hugging Face

🔍 Search models, datasets, users...



Models

Main

Tasks

Libraries

Languages

Licenses

Other

Models 2,894,455



Filter b



(I am not talking about these)

Where are all the models???

(I am not talking about these)



```
10 co = cohere.ClientV2("Your API key")
11
12 chat_dataset = co.datasets.create(
13     name="chat-dataset",
14     data=open("path/to/train.jsonl", "rb"),
15     type="chat-finetune-input",
16 )
17 # optional (define custom hyperparameters)
18 hp = Hyperparameters(
19     early_stopping_patience=10,
20     early_stopping_threshold=0.001,
21     train_batch_size=16,
22     train_epochs=1,
23     learning_rate=0.01,
24 )
25
26 # optional (define wandb configuration)
27 wnb_config = WandbConfig(
28     project="test-project",
29     api_key="<<wandbApiKey>>",
30     entity="test-entity",
31 )
32
33 create_response = co.finetuning.create_finetuned_model(
34     request=FinetunedModel(
35         name="customer-service-chat-model",
36         settings=Settings(
37             base_model=BaseModel(
38                 base_type="BASE_TYPE_CHAT",
39             ),
40             dataset_id=chat_dataset.id,
41             hyperparameters=hp,
42             wandb=wnb_config,
43         ),
44     ),
45     https://docs.cohere.com/docs/chat-fine-tuning
```

Where are all the 100B+ models?

- OpenAI 
- Anthropic
- Google 
- Meta 
- SpaceX
- Reflection 
- Poolside 

> clf = LLM(layers=70, dim=14336, seq_len=2048)

> clf.fit(pretrain = X, ift = (p,c), pref = (prompt, better, worse)))

- Mistral 
- Cohere 
- Upstage 

- Deepseek 
- Alibaba (Qwen) 
- MiniMax 
- Moonshot (Kimi) 
- Z.AI (GLM) 

Several very large companies who want their own model decide not to train in-house



Industry

- Economic Contribution / Market Recognition
- **Specialized Knowledge and Expertise**
- **Organizational Ecosystem**
- Standardization & Regulation
- Infrastructure & Investment
- Cultural & Historical Context



Ecosystem

- Annotation companies: Scale, Invisible, Turing and soooo many more
- AI orchestration companies
- Evaluation companies
- Detectors of machine-generation
- Sandbox-as-a-service

AI Agenda

The Rise of Startups That Help Other Startups Evaluate LLMs

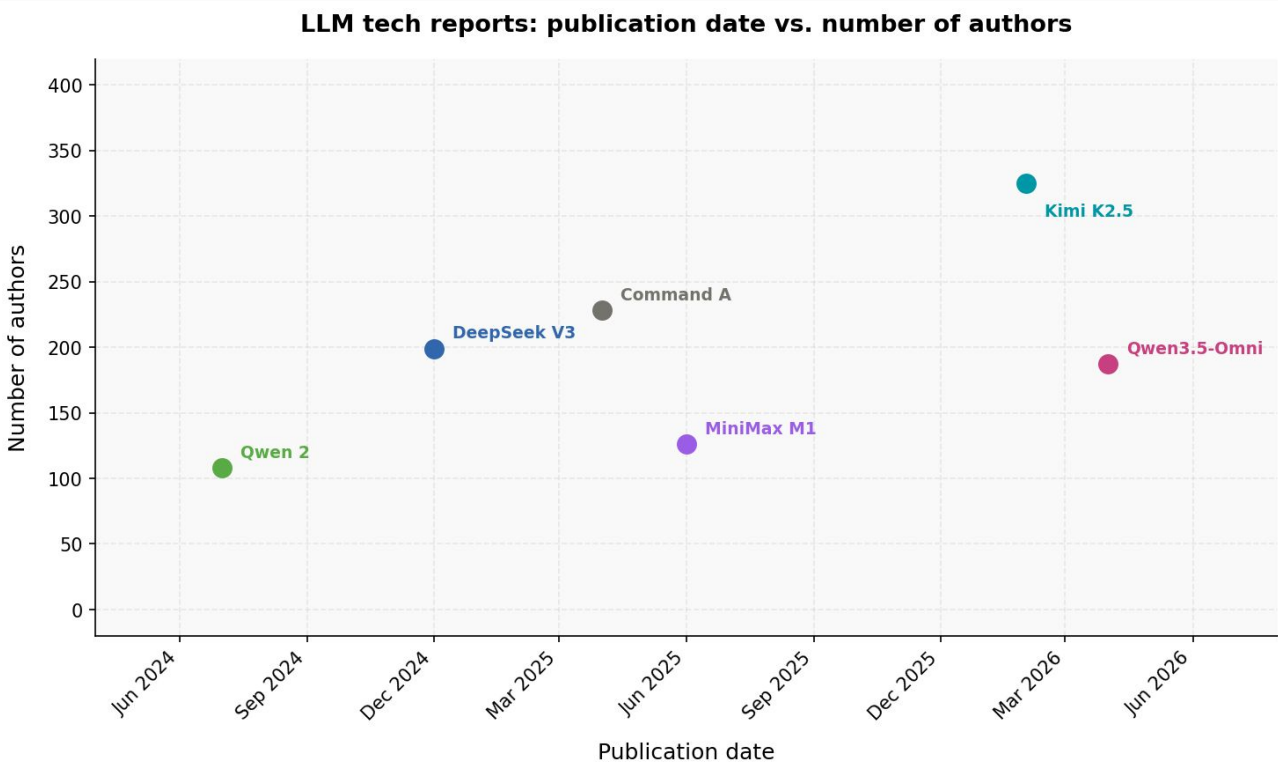
Approximating industrial scales

Paper describing Model	# authors
Deepseek v3	199
Llama 3	558
Qwen 2	108
Qwen 3.5-Omni	187
Minimax M1	126
Kimi 2.5	325
Gemini	1346
Command A	228

Paper describing Model	# authors
GPT-3	31
Olmo 2	39

Approximating industrial scales

Pa
Deep
Llam
Qwe
Qwe
Minir
Kimi
Gem
Com



authors
31
39



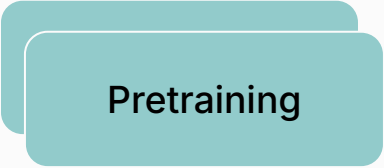
```
graph LR; A[Pretraining] --> B[IFT]; B --> C[Alignment]
```

Pretraining

IFT

Alignment

Pretraining



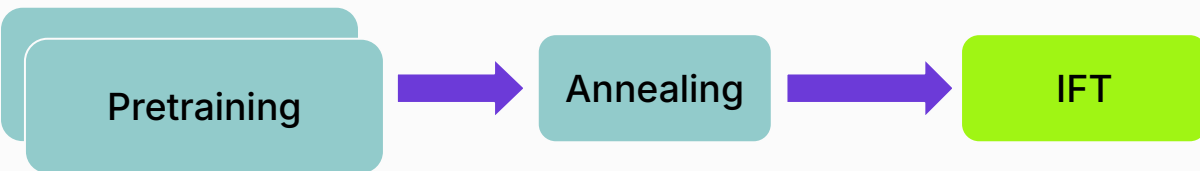
Pretraining

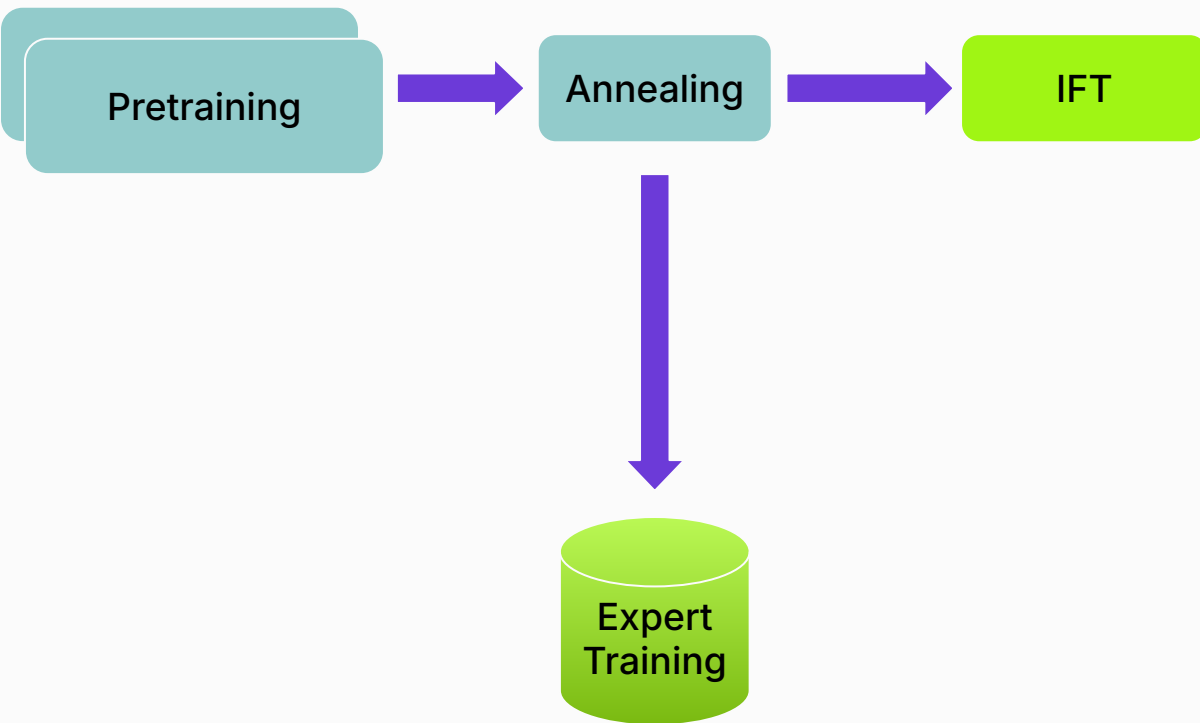


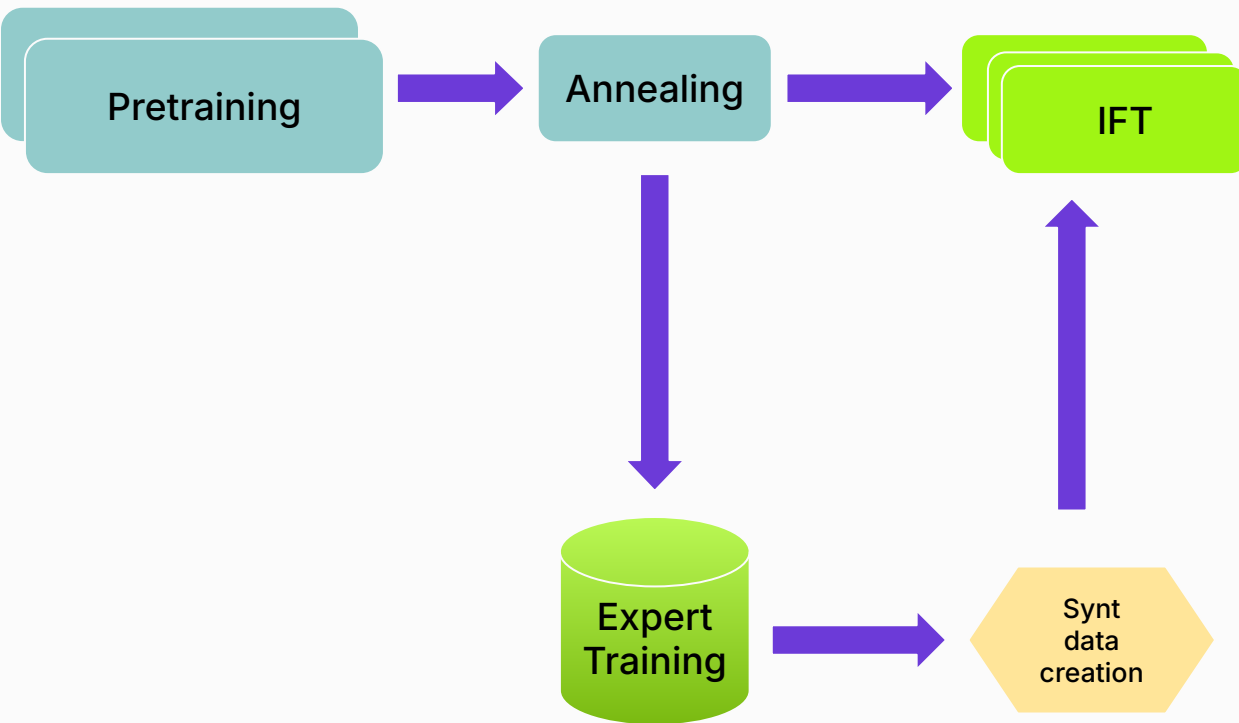
```
graph LR; A[Pretraining] --> B[Annealing]
```

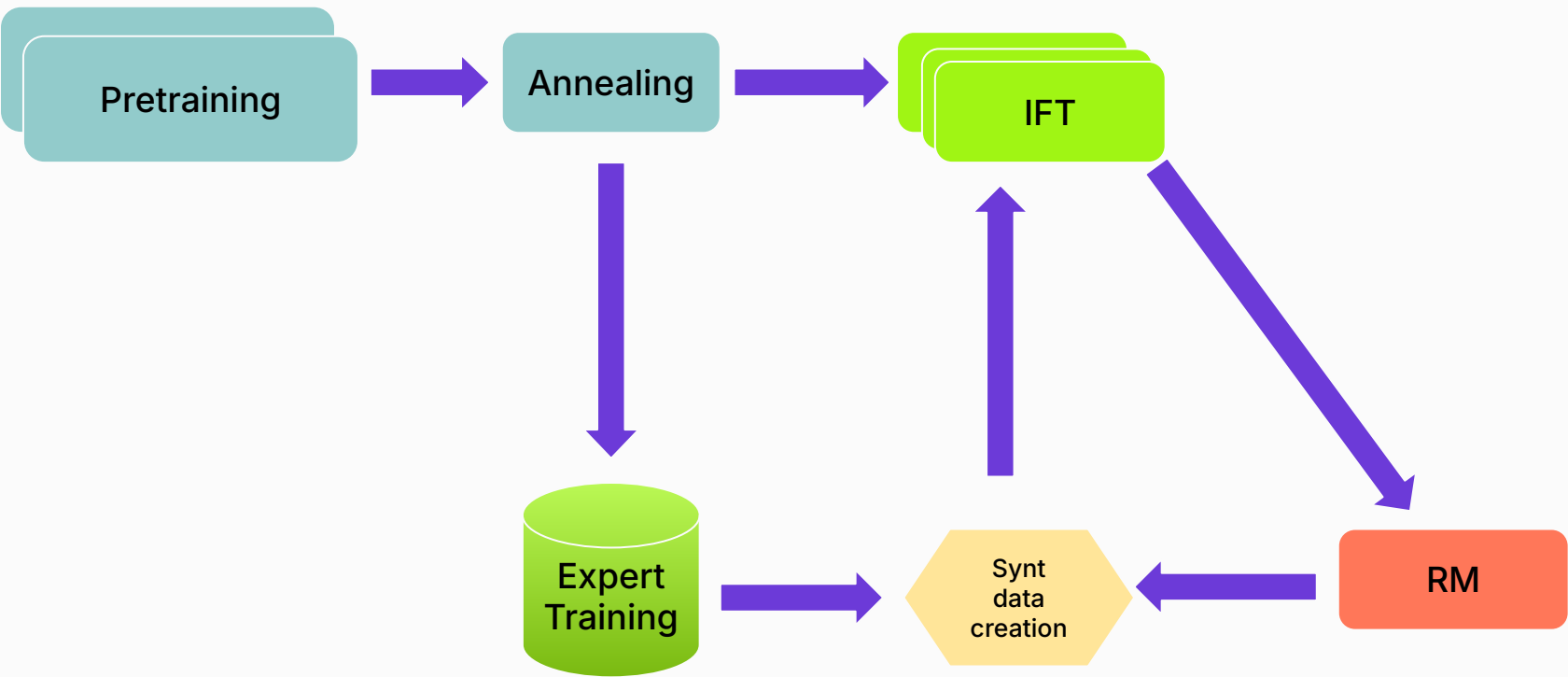
Pretraining

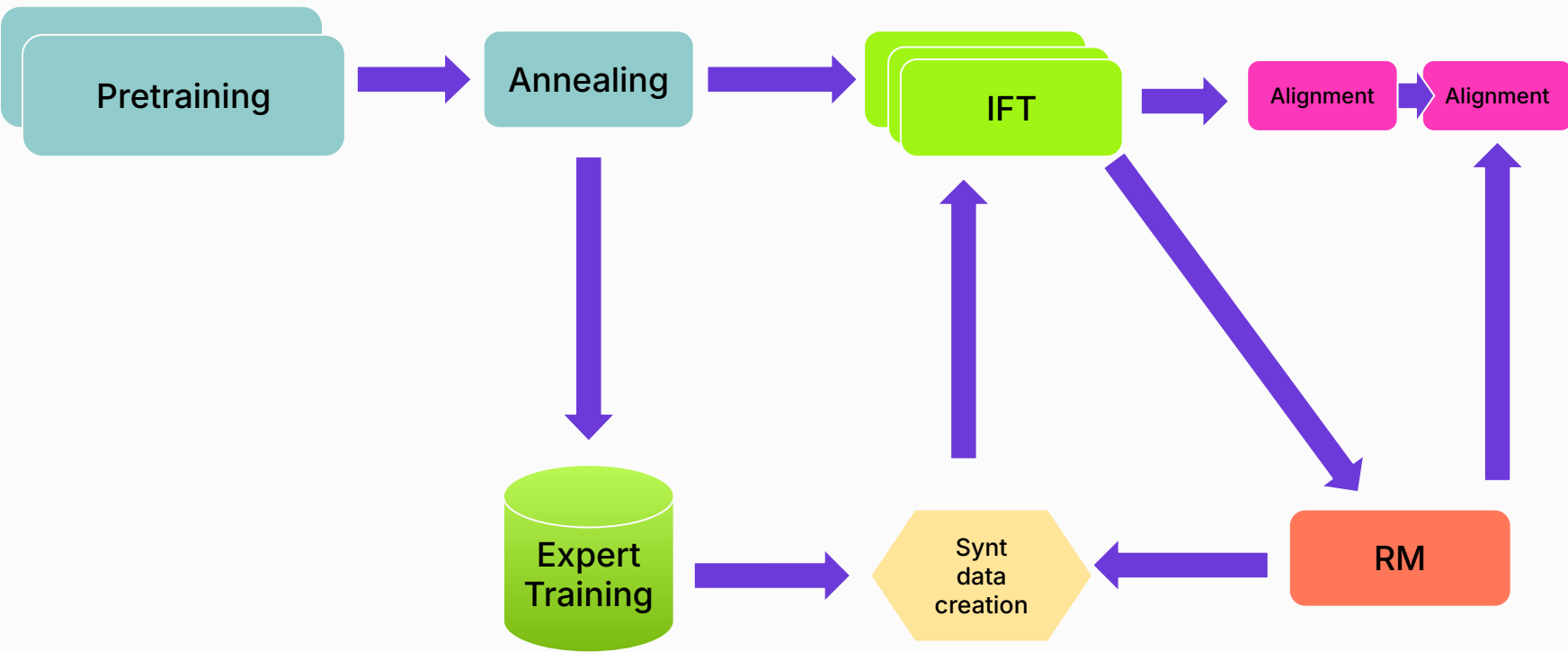
Annealing

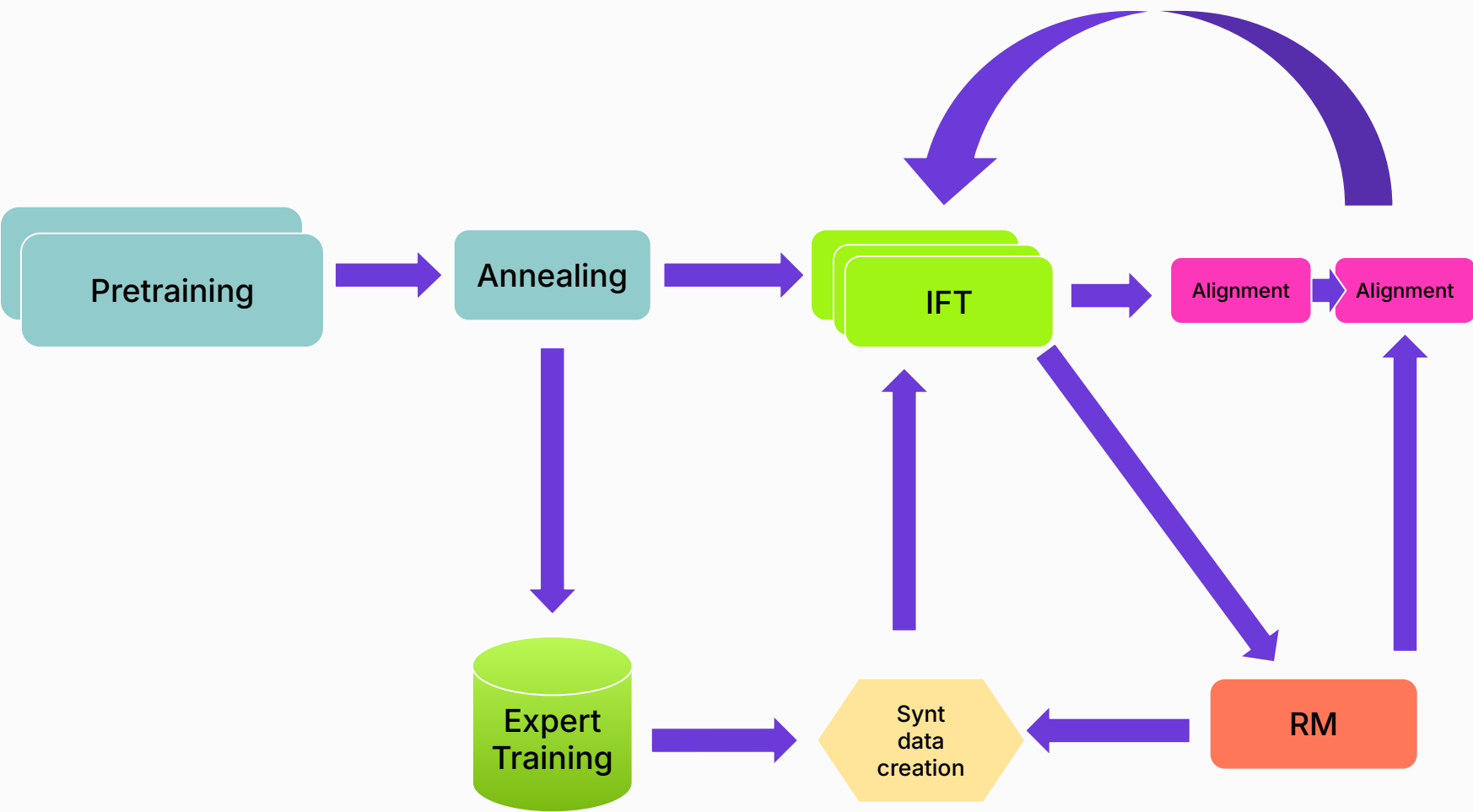


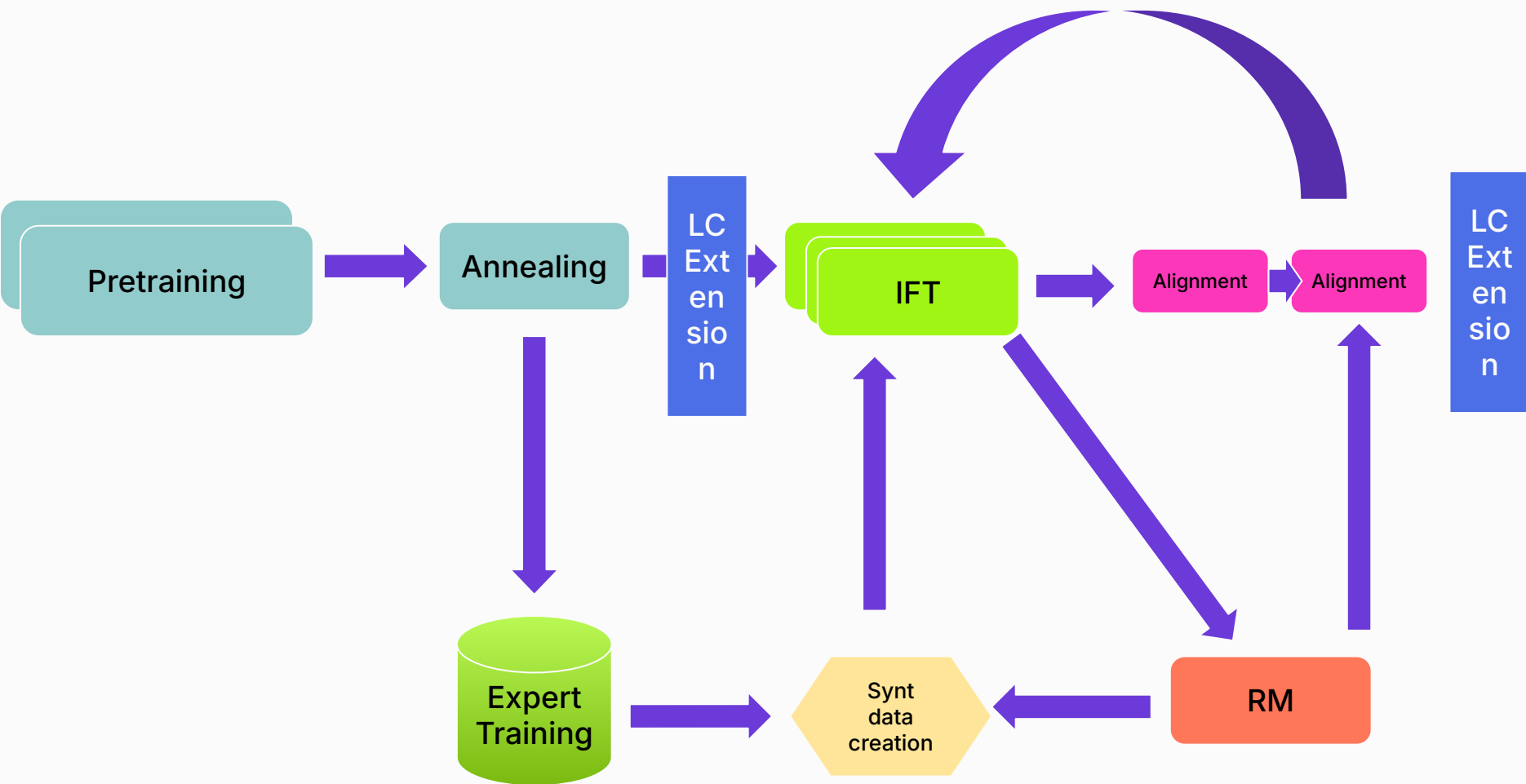




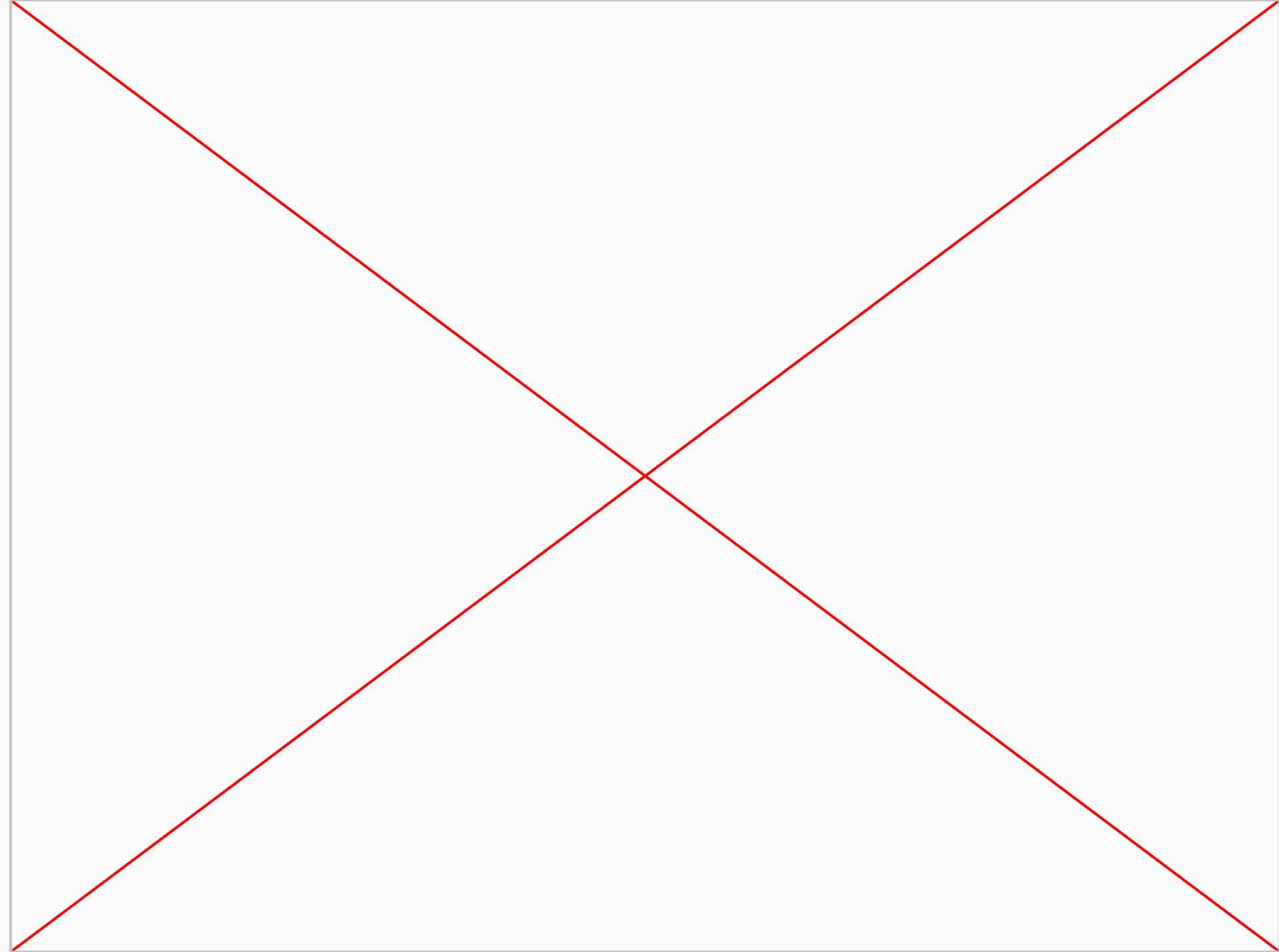


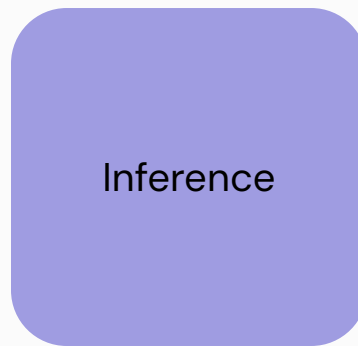
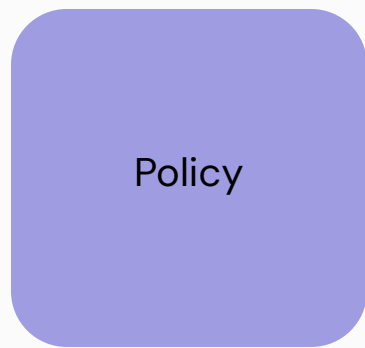






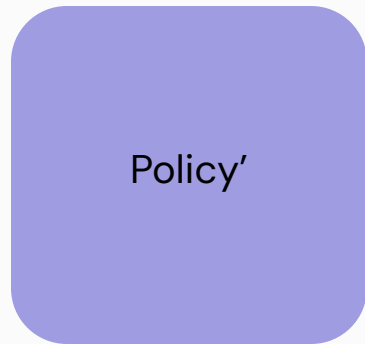
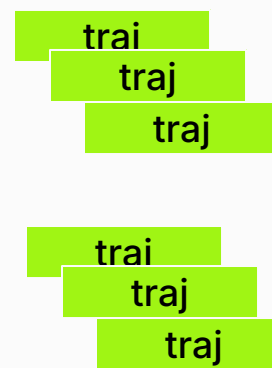
All this was pre-agentic

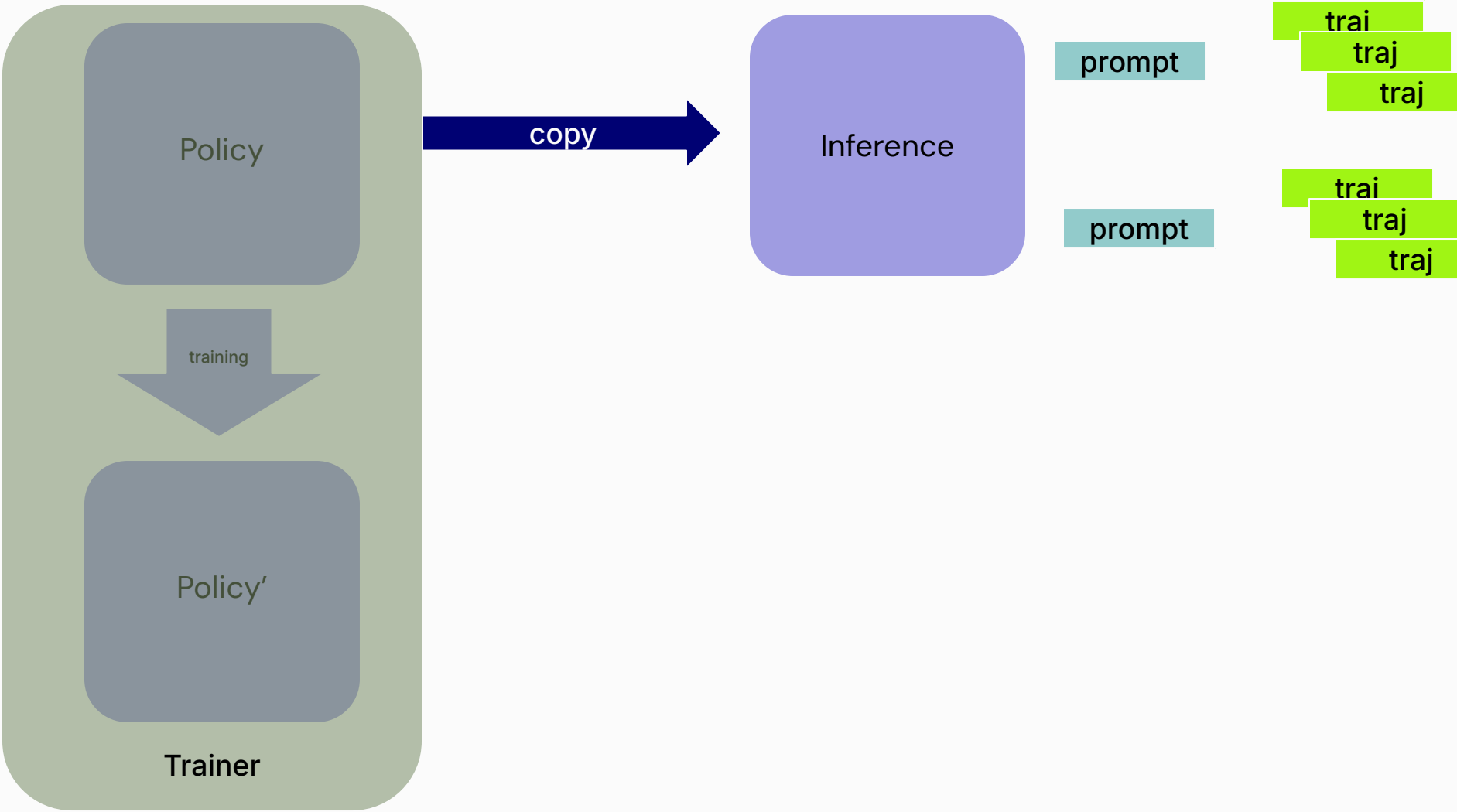


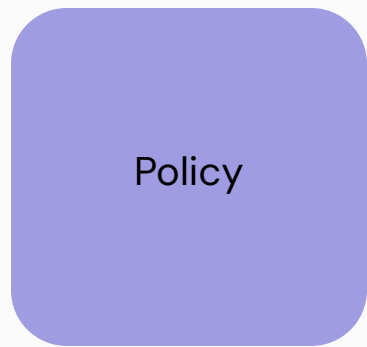


prompt

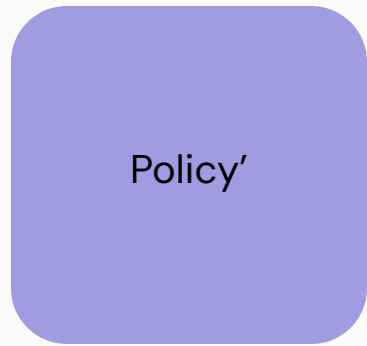
prompt







training



copy

Inference

prompt

prompt

Inference
(tp, expert parallelism, fp8 kv
cache)

trai

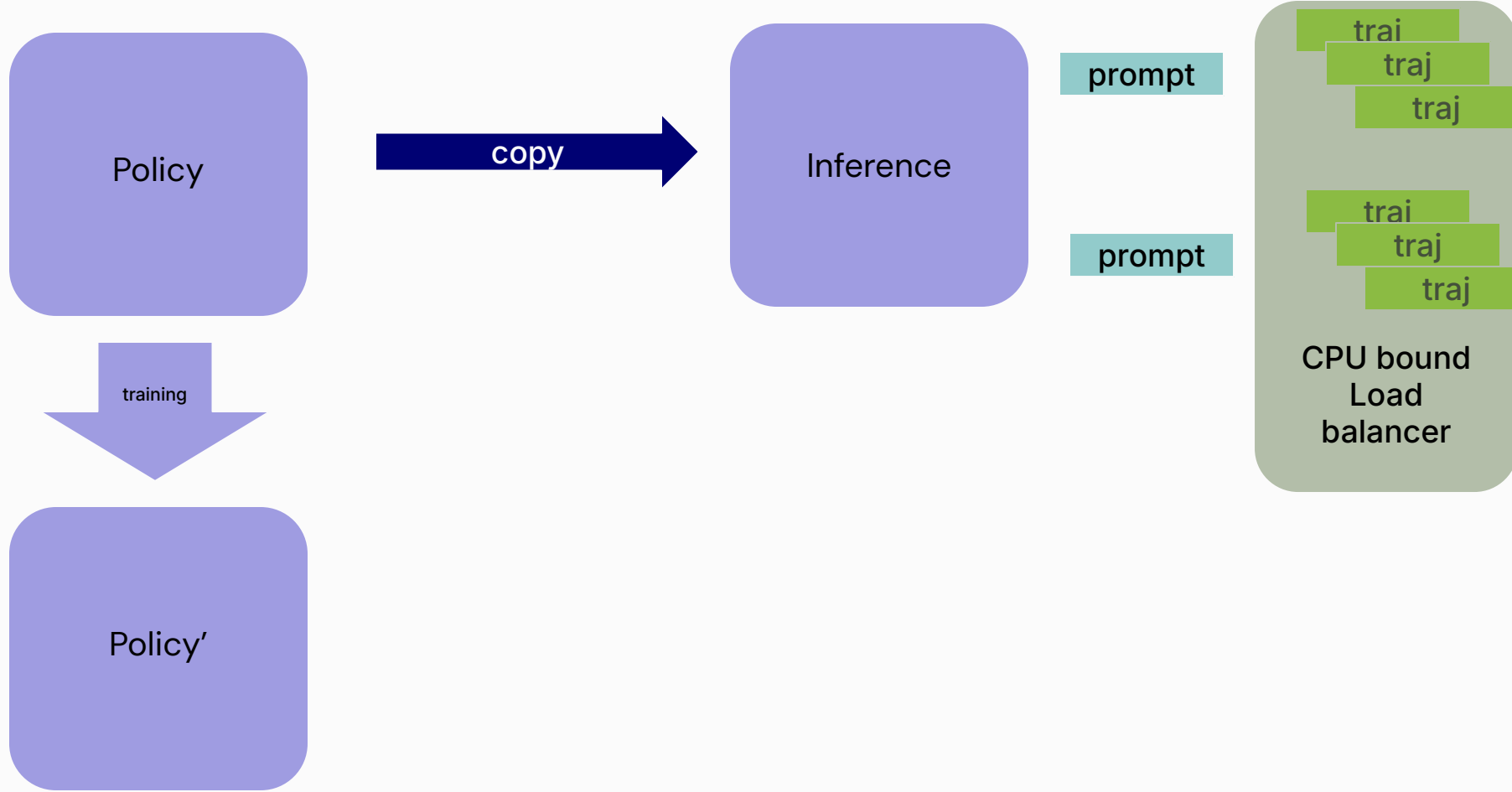
traj

traj

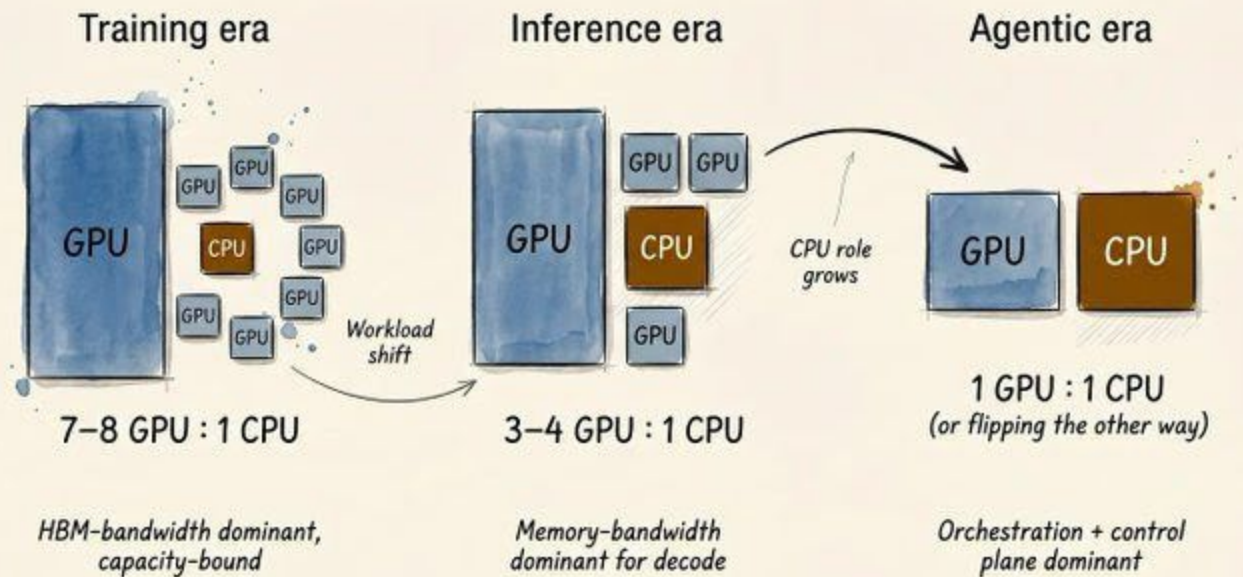
trai

traj

traj



Training vs Inference vs Agentic — GPU:CPU ratio shifts the system architecture



Source: Intel Q1 FY26 transcript, Dave Zinsner verbatim. The CPU is reasserting itself as the indispensable foundation of the AI era. — Lip-Bu Tan, Intel Q1 FY26

intel Intel Corp
NASDAQ: INTC

118,96 USD

+53.26 (81.07%) ↑ past month

Closed: 20 May, 16:59 GMT-4 • [Disclaimer](#)

After hours 118,52 -0,44 (0,37%)

1D 5D 1M 6M YTD 1Y 5Y Max



Eval surveillance

Reward hacking for agents with tool use:

- What if the environment contains the all of .git with full history?
- What if you can access github?
- What if you can access the web?

<https://poolside.ai/blog/through-the-looking-glass>

“Model Factory”

Goals:

- Reliability
- Reproducibility

To get there:

- Codify decisions as versioned assets
- Continuous evaluation
- Full data & model lineage



VS



Questions?